Information Retrieval from Tagalog and English GCash User Reviews Through Unsupervised Clustering

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***Abstract*—This study explores unsupervised learning methods on Google Play Store reviews of the GCash application, focusing on recent reviews from 2023. The study utilizes data cleaning methods, text processing with TF-IDF vectorization, and dimensionality reduction using SVD and UMAP. The research utilizes K-means and DBSCAN algorithms for clustering the text observations. K-means produced 7 clusters with a Silhouette score of 0.3794 and a DBI of 0.8309, while DBSCAN yielded 12 clusters with a Silhouette score of 0.0005 and a DBI of 0.6040. K-means showed better performance in Silhouette score and overall efficiency, whereas DBSCAN excelled in DBI, producing more distinct and interpretable clusters. The study highlights the effectiveness of both algorithms in information retrieval and clustering, despite limitations in processing Tagalog words. It suggests that future work should focus on enhanced data processing and the application of polarity identification for multilingual datasets to improve automated feedback systems, and overall improving the potential of unsupervised learning in real-world applications.**

***Index Terms*—customer review, information retrieval, natural language processing, text clustering, unsupervised learning, K-means, DBSCAN, dimensionality reduction, vectorization**

I. Introduction

The emergence of digital wallet applications in the Philippines has revolutionized the mode of payment for millions of Filipinos. Digital wallets offer cashless transactions, which are fast, secure, and a convenient method of payment and money transfer. It has even gained more traction especially during the pandemic where millions of people relied on applications like GCash to make transactions from the comfort of their homes. According to Google Play Store statistics, GCash received over 17 million downloads in 2023 [1], and over 100 million downloads at the end of 2024 [2], thus being the number one digital wallet application in the Philippines.

However, Gcash requires customer feedback to continuously improve its services and user experience. Feedback is crucial [3] for digital wallets as they handle financial transactions involving real money, consequently needing high levels of security, efficiency, and perceived trust of their user base. As of 2025, GCash [2] has gotten over 2.6 million reviews from users on the Google Play Store. This serves as an opportunity for the company to gain insights from their user base on the areas of the application that needs improvement.

The goal of this study is to determine the effectiveness of clustering techniques in analyzing user reviews, with the purpose of finding significant feedback to take into consideration for the application's improvement. To be specific, since the study would focus more on text observations, we would be able to utilize Natural Language Processing (NLP) techniques to convert the texts into a machine-readable format. Doing this, we can efficiently cluster user reviews and group each observation according to their content. In the end, this research is to contribute to the development of a data analytics approach to feedback management to support the enhancement of GCash's features and user experience.

II. Review of Related Works

Retrieving information from a vast collection of user reviews would require an exhaustive amount of effort. To efficiently extract valuable information, it is suggested to apply text clustering to group the text observations based on its word forms [4]. However, text observations first need to be refined through various NLP techniques for higher levels of granularity [5]. This refers to methods that can reduce the complexity of each word form, making it more interpretable for machine processing.

A study conducted by Murtadha, et al [11] suggests that each review must pass through proper contextual analysis before it undergoes the clustering phase. Contextual analysis includes methods such as data preparation, spelling correction, and negation handling [11]. Hence, this study would leverage contextual analysis methods for extracting important information of the text. However, certain limitations will be applied to prevent linguistic incompatibility, which will be further explained in the methodology section.

Common procedures for information extraction and text mining [6], [7] usually require implementing sentiment analysis and polarity identification to classify the opinion's tone on each text observation. The study conducted by Gourab, et al. [6] was specifically aimed to cluster user reviews according to the intensity of opinions and to identify their perspectives on the product's features, hence the clustering phase was well implemented due to the application of sentiment analysis. However, it has shortcomings on multilingual texts and text observations with prevalent code switching, even more so in informal and sarcastic reviews [8], [9] as to why application of deep neural networks are important to achieve a significant result. Unlike these approaches, our study does not perform sentiment analysis but instead focuses solely on extracting relevant information based on the clustering results.

III. Methodology

This study follows a data-driven approach to analyze and cluster user reviews, as illustrated in Figure 2.

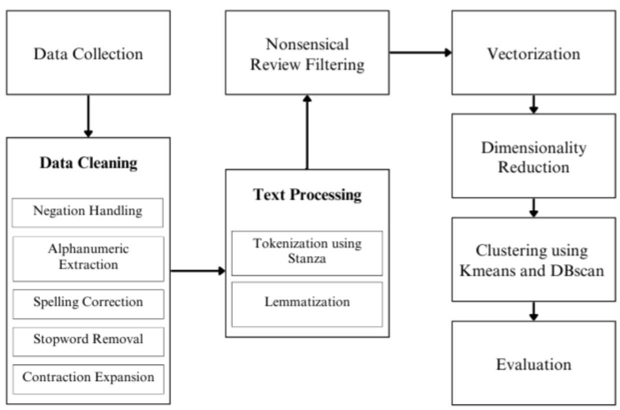

Fig. 1. GCash User Review Clustering Framework

A. Data Collection

The dataset was obtained from the Kaggle open-source database collection, containing eight columns and approximately 590,000 observations of Tagalog and English language [10]. The dataset originated from Google Play Store reviews from the year 2012 to 2023 [10]. For this study, we focus on cluster reviews from the year 2023 to extract information on the most recent volumes of text observations.

B. Data Pre-Processing

The following pre-processing techniques were applied to prepare user reviews for clustering:

1) Data Cleaning: Following the contextual analysis method by Murtadha, et al [11], several data cleaning techniques were also utilized to improve the accuracy of the model from the multilingual user review dataset.

a) Negation Handling: Negation handling [12] is a method that interprets the negation words (e.g *"no"*, *"never"*, *"couldn't"*, *"hasn't"*) and inverting the polarity of the words that are affected by negation, such as sentences like *"I cannot find the home button"* inverted to *"The home button is missing"*. This is a crucial method for NLP tasks to identify distinctive terms for each text observation. However, it poses challenges on data that has multilingual and frequent code-switching in user reviews, as to why it is necessary to implement limitations on how we can use negation handling for data cleaning [13]. The alternative that this study has sought out would be to flag the known negation words for both languages in each text observation. This should be possible to implement due to how similar the process is to the lemmatization procedure, where we can retrieve the base form of the word.

b) Alphanumeric Extraction: Alphanumeric extraction is the process of filtering out text noises that does not contribute to the textual meaning of a user review. This method is useful to extract only meaningful character and numeric elements since clustering models would not have a need for non-essential elements such as emojis, punctuation marks, and other special characters.

c) Spelling Correction: The study employs dictionary-based spelling correction mechanisms for both Tagalog and English texts, including corrections for abbreviations, internet slangs, common words used when describing the application, and informal phonetic spellings. This allows a more cleaner information retrieval and preserves the distinctive term of the corrected words.

d) Stop Words Removal: Stop words removal is a process of filtering out high-frequency but low-importance words that does not provide valuable information to the context of the text observation [14]. Common words such as "and", "is", "the" are considered stop words and should be removed to focus on the relevant context of the data.

e) Contraction Expansion: Contraction expansion [15] was implemented to convert contracted words into their full forms to enhance the text clarity. English (e.g *"can't"*, *"won't"*) and Tagalog contractions (e.g *"yung"*, *"niyo"*) were expanded to properly contextualize their intended meaning. A pre-defined dictionary of contraction-expansion mappings was initialized and used to systematically replace contractions with their expanded equivalents. Formally, let $C = C_1, C_2, C_3, \ldots, C_n$ be the set of identifiable contracted words in a sentence S. For each $C_i \in C$, there exists a corresponding expanded form $E_i \in E$, where $E = E_1, E_2, E_3, \ldots, E_m$ represents the set of available expanded words.

f) Nonsensical Review Filtering: It is common for data cleaning to exclude nonsensical text from the dataset. Standalone digits and unidentifiable characters would be excluded during data processing, as they do not contribute to the actual meaning of the user review.

2) Text Processing:

a) Tokenization: Tokenization [16] is the method of breaking down user reviews into individual words or phrases, referred to as tokens to be interpreted by the machine without losing its context. The study would fully utilize word tokenization method as a way to segment the user reviews and extract each distinctive term at a granular level.

b) Lemmatization: Lemmatization [17] is the process of reducing different morphological variations of a word to its base form or lemma. This study applies lemmatization for semantic integrity so that each word in user reviews would be able to be reduced to their root form. This improves consistency and grammatical structure after text pre-processing while maintaining the context of words that share the same meaning in their base form. Table 1 demonstrates how data cleaning and text processing were applied to extract only the relevant information in user reviews, while maintaining the actual context of the review.

Text	Text Formatted
Very frustrating app and CUSTOMER SERVICE!	['frustrating', 'application', 'customer', 'service']
I haven't registered for weeks now	['negativelabel', 'register', 'week']
Convenient for load and transfer	['convenient', 'load', 'transfer']
Very Convenient and its good to transfer transactions	['convenient', 'good', 'transfer', 'transaction']
Very helpful to pay bills online	['helpful', 'pay', 'bill', 'online']

TABLE I
DATA CLEANING AND TEXT PROCESSING RESULTS

C. Feature Extraction

a) Vectorization: To quantify the importance of a term in each text observations, we would use the Term Frequency-Inverse Document Frequency (TF-IDF), which is a scoring mechanism [11] that emphasizes distinctive terms of each text observations. TF-IDF analyzes first the term frequency of text observations and then measures the rarity of that term across the entire corpus, which is the collection of user reviews [19]. The TF-IDF is expressed mathematically through the following equation:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \tag{1}$$

$$\text{TF}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{2}$$

$$\text{IDF}(t) = \log\left(\frac{N}{|d \in D : t \in d|}\right) \tag{3}$$

where:

- t = term (word)
- d = document (a user review)
- D = collection of documents (of user reviews or corpus)
- $f_{t,d}$ = frequency of term t in document d
- N = total number of documents
- $|\{d \in D : t \in d\}|$ = number of documents containing term t

b) Dimensionality Reduction: Dimensionality reduction will be applied to reduce the number of generated features in the dataset while retaining its significant information. This process simplifies the dataset by transforming it into a lower-dimensional space [22], making it interpretable through plotting and model training without losing the feature's context. According to Avinash, the Latent Semantic Analysis (LSA) is a tool that can perform matrix decomposition, noise reduction, and convert the vector components into a low-dimensional space while maintaining the information of each feature according to frequency and importance [23]. In this study, the Singular Value Decomposition (SVD) approach will be utilized to apply LSA on text review documents and convert into low-dimensional space. Avinash stated [23] that dimensionality reduction can be applied on text data using Singular Value Decomposition (SVD). SVD [23] decomposes the original matrix into three matrices (term to topic, diagonal topic, topic to document), capturing the most important patterns while reducing noise and dimensionality in the dataset. SVD is represented mathematically through the following equation.

$$M_{m\times n} = U_{m\times n}\Sigma_{n\times n}V^T_{n\times n} \tag{4}$$

where:

- $M_{m\times m}$ = the vectorized features, where m is the number of terms (words) and n is the number of documents (a document is a user review).
- $U_{m\times n}$ = the left singular matrix, representing the term and topic semantic relationship.
- $\Sigma_{n\times n}$ = the diagonal matrix containing singular values, which indicate the importance of topic by topic.
- $V^T_{n\times m}$ = the right singular transposed matrix, representing the relationship between topics (a single topic is a group of related terms) and documents.
- m = the total number of rows in the vectorized dataset.
- n = the total number of columns in the vectorized dataset.

Following the application of SVD, the Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction technique will also be utilized to further improve computational efficiency and enable data point visualization, while preserving the SVD's dimensionality structure [24]. UMAP [24] evaluates the high-dimensionality space as a weighted graph that measures the distances of data points to each other. It will then be projected to a lower dimensional space where data points, according to the initialized nearest neighbors, are grouped together and form a cluster within their proximity, so as to preserve their original structure in the higher dimension. The general approach for the dimensionality reduction of this study follows the chained dimensionality reduction method [25], in order to optimize program efficiency and accurately use the text observations for clustering analysis.

D. Clustering Algorithms

a) K-means: The study would utilize the K-means algorithm, which is a centroid-based clustering method [18], to classify the cluster group to which the observation belongs according to the nearest initialized centroid determined by the algorithm. We first identify the number of clusters through the elbow method, then the K-means algorithm initializes centroids based on the number of clusters. Since the observations are text data of user reviews, each review should be classified according to its nearest centroid.

b) DBSCAN (Density-Based Spatial Clustering of Applications with Noise): The DBSCAN algorithm does not assume that each cluster forms a spherical shape like K-means. According to Taylor and Wilbert [21], [20], DBSCAN performs efficiently through text data and observations that possibly forms arbitrarily shaped clusters. Since GCash user reviews contains a diverse set of opinionated reviews, we conclude that using this clustering algorithm provides the study a different interpretation of clusters other than what is achieved using the K-means algorithm. This study also explored the use of agglomerative clustering as another clustering approach, however, it was proven to be extremely slow and inefficient on large datasets.

E. Experimental Setup

This study utilized Python (Jupyter Notebook in Visual Studio Code) with libraries such as NumPy, Pandas, Matplotlib, Seaborn, Regex Yellowbrick, and Scikit-Learn. Additionally, this study also utilized the Stanza library to perform word tokenization on user reviews and extracts the available lemma of every English word in the dataset. The NLTK corpus and StopwordsIso library were also utilized to find and remove every Tagalog and English stop word for data cleaning.

F. Evaluation Metrics

This study aims to identify and analyze key user feedback regarding GCash features, as revealed through information extracted from clustering analysis. However, we also intend to evaluate the research outcome based on the clustering quality. The study utilized the Silhouette Score and the Davies-Bouldin Index as secondary evaluation metrics. The Silhouette Score quantifies how well each text observation fits within its assigned cluster compared to other clusters, while the Davies-Bouldin Index measures the compactness of clusters and their separation from one another [26].

IV. Results and Discussion

A. K-means Algorithm

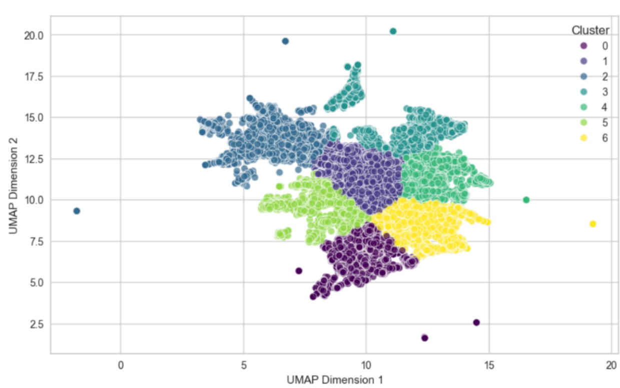

Fig. 2. 2D visualization of Kmeans clusters result

Figure 2 shows the clustering results using the Kmeans algorithm. We employ the elbow method as part of hyperparameterized tuning on the number of clusters to find the most optimized inertia score. The clustering profiling and interpretations are shown in the following tables, consisting of 7 clusters:

Word	TF-IDF Score
good	0.1055
easy	0.0954
application	0.0756
nice	0.0725
money	0.0492
use	0.0490
load	0.0461
fast	0.0423
gcash	0.0418
sometimes	0.0405

TABLE II
Top 10 words for Cluster 0

Cluster 0: General Positive User Reviews. This cluster is indicative of user satisfaction with GCash's transaction features. Complimentary words such as "good", "easy", "nice", and "fast" are among the most frequent statements in this cluster, with TF-IDF scores of *0.1055*, *0.0954*, *0.0725*, and *0.0423*, respectively. The majority of users in this cluster perceive GCash as a convenient financial tool that they can use to perform money transactions efficiently. Hence, we classify this cluster as representing general positive user reviews due to their positive sentiments toward the application.

Word	TF-IDF Score
negativelabel	0.0924
gcash	0.0621
money	0.0502
application	0.0480
please	0.0444
account	0.0431
fix	0.0383
send	0.0345
help	0.0300
open	0.0288

TABLE III
Top 10 words for Cluster 1

Cluster 1: General Negative User Reviews. We classify Cluster 1 as representing general negative user reviews due to a high TF-IDF score of *0.0924* for negation words. A high count of negation words indicates that the majority of reviews in this cluster are explicitly tied to negative sentiment. Since negative auxiliary verbs and negative adverbs are converted to "negativelabel", most of the frequent words in this cluster are accompanied by negations and describe the application's non-functionality issues, such as "couldn't send money" or "cannot open GCash".

Cluster 2: General Negative Tagalog User Reviews. Cluster 2 represents a significant portion of Tagalog user reviews, with frequent code-switching to English. The Tagalog words total a TF-IDF score of *0.4868*, and negation labels are

Word	TF-IDF Score
ninyo	0.0685
palagi	0.0642
update	0.0556
mag	0.0551
gcash	0.0532
lang	0.0442
nag	0.0394
po	0.0389
negativelabel	0.0384
pag	0.0293

TABLE IV
TOP 10 WORDS FOR CLUSTER 2

among the most frequent words in this cluster. The information provided in this cluster indicates user dissatisfaction with GCash updates. This cluster is a group of general negative Tagalog user reviews due to its association with negation words in the majority of user feedback.

Word	TF-IDF Score
verify	0.1054
negativelabel	0.0698
account	0.0621
id	0.0436
face	0.0424
gcash	0.0424
verification	0.0402
something	0.0379
wrong	0.0378
say	0.0350

TABLE V
TOP 10 WORDS FOR CLUSTER 3

Cluster 3: Verification-Related Concerns. Cluster 3 focuses on GCash security identity verification feedback. Negation words, with a TF-IDF score of *0.0698*, are further associated with the feedback, indicating user verification concerns. Words such as "verification" and "verify" are most likely associated with "face" and "account" due to high TF-IDF scores of *0.0402*, *0.1054*, *0.0424*, and *0.0621*, respectively. Therefore, we classify Cluster 3 as representing verification-related concerns.

Cluster 4: Performance and Error-Related Concerns. Words such as "slow", "open", "error", and "login" characterize this cluster. The majority of feedback reflects negative user review sentiments, with a TF-IDF score of *0.0973* for negation words. Based on the most frequent words in this cluster, we conclude that the issues are mostly related to performance degradation and error concerns regarding account login and other GCash services. Hence, this cluster centers on performance and error-related concerns.

Word	TF-IDF Score
negativelabel	0.0973
open	0.0823
login	0.0635
always	0.0597
application	0.0547
gcash	0.0510
service	0.0478
error	0.0451
slow	0.0384
account	0.0380

TABLE VI
TOP 10 WORDS FOR CLUSTER 4

Word	TF-IDF Score
application	0.0930
negativelabel	0.0778
keep	0.0503
phone	0.0489
option	0.0434
gcash	0.0397
developer	0.0388
install	0.0380
secure	0.0328
use	0.0318

TABLE VII
TOP 10 WORDS FOR CLUSTER 5

Cluster 5: Device-Related Concerns. Cluster 5 includes specific words such as "developer", "option", "device", "install", and "phone". With association to negation words, this indicates that the majority of feedback concerns device incompatibility. This study concludes that this cluster centers on device-related concerns, as it contains specific topics about GCash's restrictive features, particularly developer options and device installations.

Word	TF-IDF Score
update	0.1296
always	0.0770
application	0.0767
negativelabel	0.0590
need	0.0564
updating	0.0537
use	0.0461
time	0.0458
open	0.0434
bad	0.0345

TABLE VIII
TOP 10 WORDS FOR CLUSTER 6

Cluster 6: Update-Related Concerns. Cluster 6 is characterized by the most frequent words: “update”, “always”, “application”, “need”, “updating”, “use”, and “time”, totaling a TF-IDF score of *0.4853*. This cluster pertains to negative user experiences regarding update frequency and length. Additionally, this cluster is accompanied by “negativelabel” and shows user frustration through negation words, leading to a common structure such as: “The application can’t be used because it is always updating.”

K-Means Evaluation Score	Value
Silhouette Score	0.3794
Davies-Bouldin Index	0.8309

TABLE IX
K-MEANS CLUSTERING EVALUATION METRICS

The Kmeans clustering results achieved an overall Silhouette score of 0.3794 and a Davies-Bouldin index of *0.8309*.

The Silhouette score measures how well each data point fits within its assigned cluster. Specifically, it reflects the average Silhouette score across all data points. It is calculated by measuring a single data point’s average distance from each of the data points within its own cluster, and each of the data points that are within the nearest other cluster. Therefore, the higher the overall average of the Silhouette score, the better each data point’s assigned cluster is. The K-means cluster results achieved a Silhouette score of *0.3794*, from the minimum of -1 and the maximum of 1. This proves the clustering results achieved a moderate clustering quality. While the data points are not perfectly adjusted to their centroids, it does show meaningful structure according to the contextual cluster results of the text observations.

In addition to the Silhouette score, we evaluated the clustering results using the Davies-Bouldin Index (DBI), which yielded a score of *0.8309*. The Davies-Bouldin Index measures the average distance of all the data points within a cluster to its own centroid. Additionally, it also measures the distance of the entire cluster from the nearest neighboring cluster to determine how compact and separated each cluster is within the dataset. The higher the DBI yields, the more spread out each of the data points are within their own clusters and the clusters are closer to each other. The DBI score of *0.8309* indicates a good cluster separation and data point compaction, resulting in spatially distinct clusters.

B. DBSCAN Algorithm

Figure 3 shows the clustering results using the DBSCAN algorithm. The DBSCAN clustering results are shown in the following tables, consisting of 12 clusters:

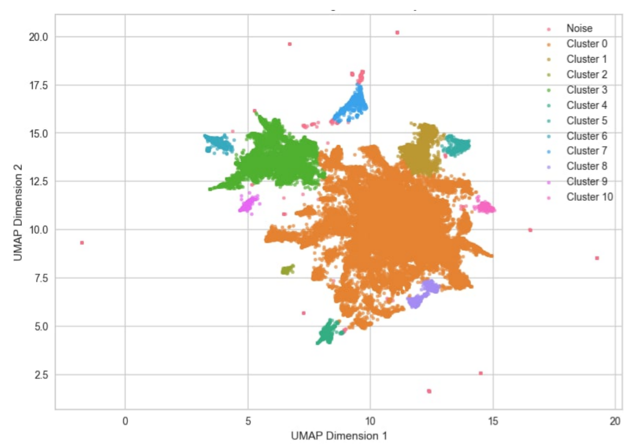

Fig. 3. 2D visualization of DBSCAN clusters result

Word	TF-IDF Score
negativelabel	0.0735
application	0.0655
gcash	0.0485
update	0.0457
open	0.0385
always	0.0371
use	0.0308
money	0.0271
need	0.0261
good	0.0259

TABLE X
TOP 10 WORDS FOR CLUSTER 0

Cluster 0: General Reviews. This cluster represents the majority of reviews and highlights users’ frustration with GCash’s application performance. Keywords such as “negativelabel”, “application”, “update”, and “open” are among the most frequent terms in this cluster, with TF-IDF scores of 0.0735, 0.0655, 0.0457, and 0.0385, respectively. The majority of users in this cluster suggests issues related with app stability and accessibility, particularly login difficulties and persistent problems with updates, suggesting that users expect a more seamless experience when using the app.

Cluster 1: Account and Verification Reviews. This cluster reflects widespread dissatisfaction with GCash’s verification process. Keywords such as “verify”, “account”, “id”, and “verification” are among the most frequent terms in this cluster, with TF-IDF scores of 0.2169, 0.0991, 0.0897, and 0.0662, respectively. The majority of users in this cluster report struggles with long processing times and difficulties in fully verifying their accounts, with complaints about waiting periods ranging from days to months, suggesting frustration over the prolonged verification process.

Word	TF-IDF Score
verify	0.2169
account	0.0991
id	0.0897
negativelabel	0.0896
verification	0.0662
still	0.0515
fully	0.0507
day	0.0393
take	0.0376
month	0.0373

TABLE XI
TOP 10 WORDS FOR CLUSTER 1

Word	TF-IDF Score
download	0.4551
negativelabel	0.1268
application	0.1146
gcash	0.0545
hate	0.0377
uninstall	0.0354
long	0.0311
version	0.0305
update	0.0298
matagal	0.0262

TABLE XII
TOP 10 WORDS FOR CLUSTER 2

Cluster 2: Application Download and Update Reviews. This cluster represents complaints about app downloads and updates. Keywords such as “download”, “uninstall”, “version”, and “update” are among the most frequent terms in this cluster, with TF-IDF scores of 0.4551, 0.0354, 0.0305, and 0.0298, respectively. The majority of users in this cluster experience problems installing the application and report dissatisfaction with app performance after updates, including slow response times and instability.

Cluster 3: Tagalog Language and Application Reviews. This cluster primarily consists of complaints written in Tagalog, emphasizing dissatisfaction with frequent updates. Keywords such as “update”, “ninyo”, “palagi”, and “puro” are among the most frequent terms in this cluster, with TF-IDF scores of 0.0625, 0.0780, 0.0743, and 0.0302, respectively. The majority of users in this cluster believe that updates are excessive and disrupt app functionality, with additional concerns about slow performance and a desire for better optimization.

Word	TF-IDF Score
ninyo	0.0780
palagi	0.0743
update	0.0625
mag	0.0606
lang	0.0487
gcash	0.0466
nag	0.0447
negativelabel	0.0377
nalang	0.0329
pag	0.0322

TABLE XIII
TOP 10 WORDS FOR CLUSTER 3

Word	TF-IDF Score
nice	0.5166
application	0.1107
money	0.0504
easy	0.0493
good	0.0454
use	0.0383
gcash	0.0355
verry	0.0301
transaction	0.0256
send	0.0196

TABLE XIV
TOP 10 WORDS FOR CLUSTER 4

Cluster 4: Positive Application Experience Reviews. This cluster indicates user satisfaction with GCash’s services. Keywords such as “nice”, “good”, “easy”, and “fast” are among the most frequent terms in this cluster, with TF-IDF scores of 0.5166, 0.0454, 0.0493, and 0.0177, respectively. The majority of users in this cluster perceive GCash as a convenient financial tool for transactions, with additional mentions of security and helpfulness reinforcing trust in the app.

Word	TF-IDF Score
face	0.3162
recognition	0.1444
scan	0.1257
negativelabel	0.0849
verification	0.0771
open	0.0477
account	0.0467
login	0.0449
remove	0.0442
gcash	0.0425

TABLE XV
TOP 10 WORDS FOR CLUSTER 5

Cluster 5: Face Recognition and Login Reviews. This cluster focuses on problems related to facial recognition for account access. Keywords such as "face", "recognition", "scan", and "fail" are among the most frequent terms in this cluster, with TF-IDF scores of 0.3162, 0.1444, 0.1257, and 0.0319, respectively. The majority of users in this cluster struggle with this verification method, with some expressing a preference for alternative authentication options.

Word	TF-IDF Score
po	0.3692
gcash	0.0813
negativelabel	0.0409
paki	0.0372
ninyo	0.0370
sana	0.0337
maopen	0.0327
mag	0.0295
open	0.0278
ayos	0.0269

TABLE XVI
TOP 10 WORDS FOR CLUSTER 6

Cluster 6: Tagalog Support Requests and Application Functionality Reviews. This cluster features feedback in Tagalog, highlighting various user concerns. Keywords such as "po", "paki", "ayos", and "maopen" are among the most frequent terms in this cluster, with TF-IDF scores of 0.3692, 0.0372, 0.0269, and 0.0327, respectively. The majority of users in this cluster request fixes for issues such as difficulty opening the app and performance problems after updates.

Word	TF-IDF Score
pag	0.0113
mo	0.0088
lang	0.0046
ok	0.0046
tapos	0.0040
sana	0.0040
datum	0.0039
mag	0.0039
ninyo	0.0038
hassle	0.0037

TABLE XVII
TOP 10 WORDS FOR CLUSTER 7

Cluster 7: Mixed Tagalog Feedback and General Usage Reviews. This cluster contains a mix of user complaints without a dominant issue. Keywords such as "ok", "hassle", and "system" are among the most frequent terms in this cluster, with TF-IDF scores of 0.0046, 0.0037, and 0.0031, respectively. The majority of users in this cluster express general dissatisfaction, but the relatively low TF-IDF scores suggest that this cluster captures diverse but less frequent concerns.

Word	TF-IDF Score
loading	0.2327
sometimes	0.2325
open	0.0694
negativelabel	0.0639
application	0.0605
slow	0.0559
always	0.0476
good	0.0467
update	0.0392
keep	0.0273

TABLE XVIII
TOP 10 WORDS FOR CLUSTER 8

Cluster 8: Loading and Performance Reviews. This cluster highlights frustration with app speed and responsiveness. Keywords such as "loading", "sometimes", "slow", and "open" are among the most frequent terms in this cluster, with TF-IDF scores of 0.2327, 0.2325, 0.0559, and 0.0694, respectively. The majority of users in this cluster report encountering lag and difficulties opening the app, indicating significant performance issues.

Word	TF-IDF Score
ok	0.3884
sana	0.0424
gcash	0.0421
lng	0.0411
lang	0.0363
negativelabel	0.0312
and	0.0299
gamitin	0.0298
kaso	0.0269
update	0.0251

TABLE XIX
TOP 10 WORDS FOR CLUSTER 9

Cluster 9: Mixed Feedback and Tagalog Terms. This cluster consists of brief reviews expressing both positive and negative sentiments. Keywords such as "ok", "sana", "good", and "update" are among the most frequent terms in this cluster, with TF-IDF scores of 0.3884, 0.0424, 0.0201, and 0.0251, respectively. The majority of users in this cluster mention app performance and usability, often in short phrases.

Word	TF-IDF Score
safe	0.4092
money	0.1160
negativelabel	0.1123
anymore	0.0866
gcash	0.0542
application	0.0533
good	0.0388
use	0.0358
security	0.0276
lot	0.0248

TABLE XX
TOP 10 WORDS FOR CLUSTER 10

Cluster 10: Security and Safety Reviews. This cluster contains feedback about security-related concerns. Keywords such as "safe", "negativelabel", "anymore", and "scam" are among the most frequent terms in this cluster, with TF-IDF scores of 0.4092, 0.1123, 0.0866, and 0.0159, respectively. The majority of users in this cluster appreciate GCash's security features but also express concerns about trust, lost funds, and scams.

Word	TF-IDF Score
maintenance	0.3883
always	0.1315
negativelabel	0.0599
application	0.0354
need	0.0337
time	0.0283
gcash	0.0240
kayo	0.0237
go	0.0204
emergency	0.0201

TABLE XXI
TOP 10 WORDS FOR CLUSTER 11

Cluster 11: Maintenance and Service Disruptions Reviews. This cluster focuses on frustration with app maintenance. Keywords such as "maintenance", "always", "need", and "time" are among the most frequent terms in this cluster, with TF-IDF scores of 0.3883, 0.1315, 0.0337, and 0.0283, respectively. The majority of users in this cluster complain about frequent maintenance and its impact on accessibility, especially during emergencies.

K-Means Evaluation Score	**Value**
Silhouette Score	0.0005
Davies-Bouldin Index	0.6040

TABLE XXII
DBSCAN CLUSTERING EVALUATION METRICS

The DBSCAN clustering results yielded a Silhouette score of *0.0005* and a Davies-Bouldin Index (DBI) of *0.6040*.

The Silhouette score of *0.0005* indicates that data points are poorly adjusted to their respective centroids. However, this result does not indicate an uninterpretable observations. The contextual results provides meaningful results, as such review profiling could be efficiently performed despite the low Silhouette score.

The DBI on the other hand, yielded a score of *0.6040*. This indicates a highly compacted data points and providing spatially distinct clusters. This is a noticeable improvement compared to the K-means clustering results.

C. Clustering Algorithms Comparison

According to the metric results, the K-means algorithm yielded a higher Silhouette score than the DBSCAN algorithm. However, the DBSCAN algorithm outperformed K-means in terms of the Davies-Bouldin Index (DBI). This suggests that while K-means exhibited a stronger relationship between centroids and data points, DBSCAN achieved better data point compactness and cluster separability.

In this study, both the Kmeans and DBSCAN algorithm accurately clustered user reviews that are centered around the general positive and negative sentiments. However, the characteristic variation per cluster are different. Such as DBSCANS accurately grouping intensified and time-based user statements, while Kmeans grouping most of the Tagalog user reviews. The DBSCAN shows its capability to group highly specific user reviews, at the cost of system performance, while Kmeans offered a generalized approach while being incapable of handling arbitrary shaped data. Though, it offers a more stable and efficient clustering approach. We believe that both of algorithms offer distinct use-cases; however for this specific dataset, Kmeans outperformed DBSCAN in terms of efficiency and metric score.

V. CONCLUSION

In this paper, we have explored the effectiveness of unsupervised learning methods for information retrieval on Google Play Store reviews of the GCash digital wallet application. Several data cleaning and text processing techniques were implemented to perform information retrieval on each text observation. TF-IDF was used to transpose text observations into numerical vectors. To optimize the cluster results, SVD and UMAP were employed to reduce the dimensionality of the vectorized text features. The K-means algorithm produced 7 clusters, while DBSCAN yielded 12. Both clustering algorithms were profiled based on the text observations within each cluster. K-means achieved a Silhouette score of 0.3794 and a DBI of 0.8309. DBSCAN, on the other hand, yielded a Silhouette score of 0.0005 and a DBI of 0.6040. K-means outperformed DBSCAN in terms of the Silhouette score, while DBSCAN yielded a better DBI score. Overall, K-means outperformed DBSCAN in terms of evaluation metrics and efficiency. However, the DBSCAN cluster results were more distinct, interpretable, and highly specific based on the most frequent words of each cluster.

Given that the focus of this study was information retrieval and clustering results, we strongly recommend potential improvements in data cleaning and text processing to optimize cluster interpretation. Several limitations were encountered due to the difficulty of cleaning and processing most of the Tagalog words. Future studies could benefit from our data preparation approach for text clustering. We believe that improvements such as the application of polarity identification on this multilingual dataset can yield more optimized empirical results in order to apply this study in real-world scenarios, such as the development of effective automated feedback management systems utilizing unsupervised learning techniques.

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